

5TH INTERNATIONAL CONFERENCE

Smart Grid Synchronized Measurements & *Analytics*

Benchmarking Dynamic Frequency Estimators for Low-Inertia IBR Grids

Jorge L. Mayorga Taborda, Yessica Africano Rodriguez, Fernando Jimenez

Universidad de los Andes | SGSMA 2026



Santiago de Chile · June 1-4, 2026

Date

June 1-4, 2026

Venue

Santiago, Chile



Slide 1 / 60 ~30s cum 0:30

CORE

Title

Say:

- Good morning, everyone. I am Jorge Mayorga, from Universidad de los Andes.
- I want to open with a deceptively simple question: in a grid run by inverters, which frequency estimator should we actually trust?
- The honest answer is, it depends, and the goal of this talk is to make 'it depends' precise and useful.

→ *So let me start with why this is a problem worth solving.*

Key line: There is no universal best estimator; the right choice depends on what you measure.

Motivation (section)

Key line: Why this matters.

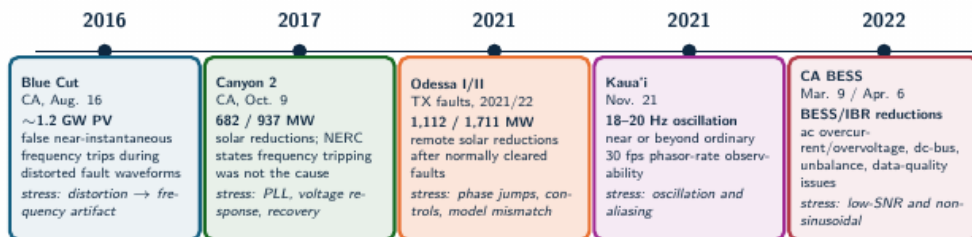
Motivation

Why frequency estimation in low-inertia IBR grids is hard

SGSMA 2026 · SANTIAGO DE CHILE



Field events define the benchmark stress model



Benchmark implication: the estimator must be tested against waveform distortion, phase jumps, control recovery, unbalance, noise, harmonics, interharmonics, and reporting-rate limits, not only steady-state frequency error.

Source: NERC Blue Cut 2017; NERC/WECC Canyon 2 2018; NERC Odessa 2021/2022; NERC/WECC CA BESS 2023; Dong et al. Kaua'i analysis

Field events define the stress model

Say:

- Real inverter disturbances look nothing like a clean sine wave.
- In events like Blue Cut, Odessa, and the Kauai oscillation, phase jumps, fast RoCoF, harmonics and noise all arrive at once.
- So instead of inventing test signals, we let these real events define the stress cases the benchmark must cover.

→ And here is the catch: the very tool we use to watch these events also shapes what we conclude.

Key line: Field events, not textbook signals, set the bar.

PMU streams show consequences

Say:

- A standard PMU hands us frequency and RoCoF.
- But during an inverter event those numbers show the consequence, not always the cause, and it is the estimator inside the PMU that decides what we actually see.

→ So before we can trust any analysis of an event, we have to ask which estimator produced it.

Key line: Choosing the estimator is itself a measurement decision.

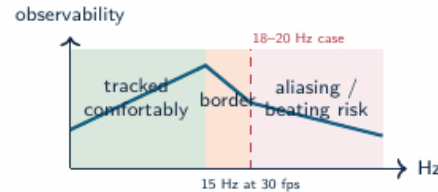
Standard PMU streams show consequences, not always causes

Fast fault and ride-through

IBR controls can react within cycles. A low-rate or heavily filtered frequency estimate may miss the subcycle cause while still showing the delayed consequence.

Oscillation near reporting-rate limits

The Kaua'i Island event on Nov. 21, 2021 exhibited 18–20 Hz oscillations after an oil-power-plant trip. A 30 frame/s phasor stream has a 15 Hz Nyquist limit for phasor-time variations, so oscillations near this range can alias or appear as beating.



Harmonics and interharmonics

Standard synchrophasor outputs summarize the fundamental component; harmonic and interharmonic contamination can still dominate estimator failure modes.



Research problem

- Low-inertia IBR grids require reliable local frequency estimates during ramps, phase jumps, harmonic distortion, ringdown, and mixed dynamic events.
- Classical estimators often trade off responsiveness, noise rejection, latency, and computational load.
- Modern model-based and data-driven estimators improve some regimes, but can be difficult to compare fairly.
- Existing comparisons often under-specify seeds, disturbance families, tuning policy, trip-risk, structural latency, or computational cost.

Research gap

The literature needs a reproducible benchmark that connects estimator error to protection-oriented risk, settling behavior, and deployability.

Say:

- And that exposes a real gap: today there is no agreed way to compare these estimators under inverter-specific stress.
- Compliance tests run on clean signals, so they simply never tell us what happens during a real composite event.

→ *Our response to that gap is one clear claim.*

Key line: We need a fair, inverter-relevant way to compare estimators.



Operating claim

- Frequency estimation under grid disturbances is a **multi-objective measurement problem**.
- Rankings change with the objective: frequency RMSE, RoCoF error, latency, CPU cost, and disturbance sensitivity produce different orders.
- IBR-era disturbances make this operational: distorted waveforms and fast controls can turn a measurement artifact into a ride-through or forensic diagnosis problem.
- Estimator choice is conditional: event family, objective metric, and compute budget determine the preferred method.

Measurement rule

Report error, transient exposure, settling, latency, and cost as separate outputs for each estimator.

Slide 6 / 60 ~25s cum 2:34

CORE

Operating claim

Say:

- The claim is this: estimator selection has to be conditional.
- It depends on the disturbance type, on the metric you care about, and on your compute budget, all three together.

→ *And to back that claim, we built three things.*

Key line: There is no single best estimator, only a best one for a given job.



Technical contributions

Not claimed here

- PMU time-stamp certification;
- a single universal leaderboard;
- hardware latency measurements.

Contributions

- OpenFreqBench: a reproducible stress-test benchmark and platform for relay-class estimators;
- a metric-complete comparison protocol: RMSE, peak error, RoCoF error, T_{trip} , settling, and CPU;
- evidence that standard compliance tests miss IBR failure modes;
- an adapted RA-EKF as a tested estimator, with explicit $\hat{\omega}$ state and event gating.

The benchmark explains why estimator rankings change across objectives, regimes, and compute budgets.

Technical contributions

Say:

- First, an open benchmark platform with known-truth scenarios.
- Second, a common execution contract, so all eighteen estimators are tested under identical conditions.
- And third, a metric vector that keeps accuracy, trip-risk, latency and cost as separate outputs.

→ *Let me give you a feel for the scale of this.*

Key line: Open platform, common contract, multi-metric evidence.

Experiment scale

18
estimators

5
scenarios

4
families

grid
search

Central finding

No single estimator wins across objectives: the ranking changes with scenario, metric, and compute budget. Protection-class frequency measurement needs tests beyond compliance-style cases.

Dataset: 18 estimators (incl. legacy baselines) across 33 scenarios, $n = 5$ Monte Carlo per scenario, with standard and IBR-specific metrics, plus ATLAS severity sweeps.

Experiment scale

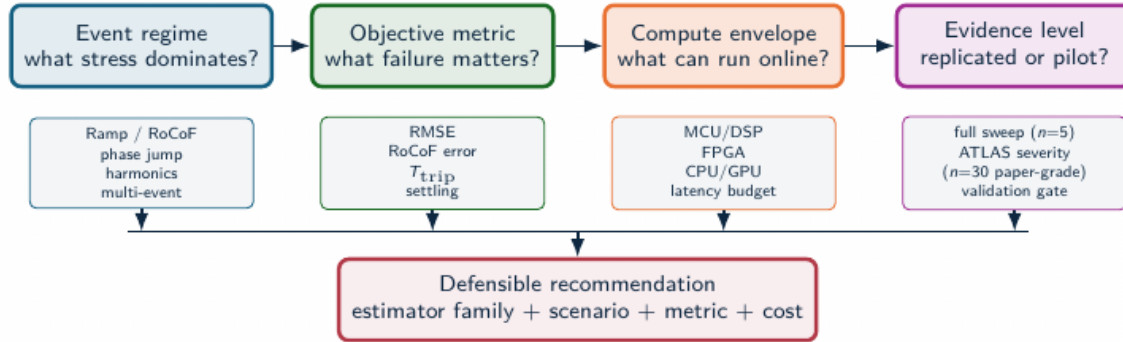
Say:

- Eighteen estimators, thirty-three scenarios, five families of stress, all with Monte-Carlo repetitions.
- And on top, a severity-sweep engine we call ATLAS, with everything reproducible from saved artifacts.

→ *The logic for choosing among all of these actually fits in a single picture.*

Key line: Large, factorial, and fully reproducible.

Estimator selection logic



Example: RA-EKF is recommended for ramp/phase-jump protection metrics under DSP/FPGA-class cost; global RMSE leadership remains scenario-dependent.

Slide 9 / 60 ~30s cum 3:59

FIG

Estimator selection logic (build-up)

Say:

- It comes down to three questions: which event, which metric, and which compute budget.
- Watch as each one appears, because each one can point you to a different method.

→ *With that logic in hand, let me show you the platform itself.*

Key line: This one diagram is the whole talk in miniature.

Benchmark and platform (section)

Key line: First, the benchmark and the platform.

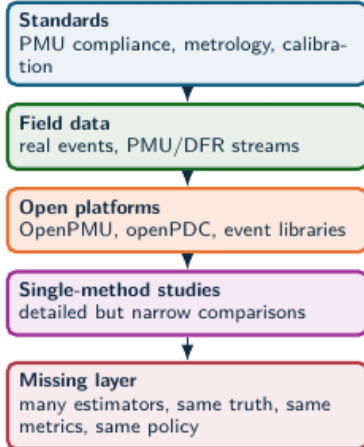
Benchmark & platform

What is missing in prior work — and OpenFreqBench

SGSMA 2026 · SANTIAGO DE CHILE



Benchmark literature gap



Why existing evidence is not enough

Standards define requirements; field events define realism; open PMU ecosystems provide infrastructure. None alone answers which estimator family is defensible under the same disturbance, tuning policy, metric vector, and compute budget.



Estimator-level benchmark: one artifact contract for algorithms, scenarios, seeds, metrics, and plots

Result: controlled synthetic truth isolates estimator behavior under reproducible stress while field events define realism.

Benchmark literature gap

Say:

- Most prior work compares a handful of estimators on one or two cases.
- What is missing is a benchmark that covers the full estimator space against composite inverter stress, with reproducible metrics.

→ *So that is precisely what we built.*

Key line: That blank space on the map is the gap we fill.

Benchmark comparison axes

Evidence source	Known truth	IBR composite stress	Trip-risk	CPU/latency	Reproducible artifacts
IEC/IEEE-style compliance	Yes	Limited	No	No	Device-dependent
Field PMU/DFR data	No	Yes	Possible	No	Dataset-dependent
Single-method studies	Often	Partial	Partial	Partial	Often limited
Estimator benchmark	Yes	Yes	Yes	Yes	Yes

Measured outputs

The same voltage trace is scored by RMSE, peak error, RoCoF error, trip exposure, settling, CPU, and structural latency.

Interpretive effect

An estimator can pass standard cases and still fail a composite IBR event; it can also be accurate but too delayed or costly for protection-adjacent use.

Benchmark comparison axes

Say:

- We compare on four axes at the same time: accuracy, robustness, computational cost, and protection trip exposure.
- And the whole point is that a method can win on one axis and quietly lose on another.

→ *Here is the platform that measures all four.*

Key line: One number hides the trade-off; four axes reveal it.

OpenFreqBench platform

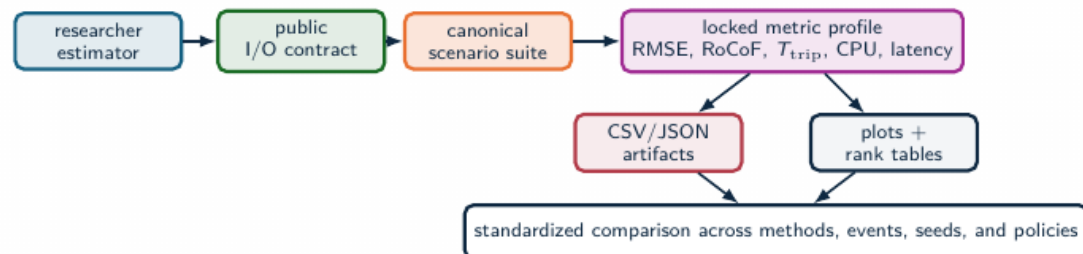
Say:

- OpenFreqBench has four parts: a scenario factory with known truth, a uniform estimator interface, a locked metric engine, and fully traceable artifacts.

→ Let me show you exactly how we keep the comparison fair.

Key line: Known truth in, locked metrics out, everything traced.

OpenFreqBench platform



Researchers add

Estimator code, parameters, and hypotheses through the public contract.

Platform fixes

Truth signals, sampling, seeds, tuning policy, metric formulas, and schema.

Outputs

Benchmark reports, metric tables, plots, manifests, and readiness gates.

OpenFreqBench turns a new estimator into a repeatable benchmark entry instead of a one-off comparison.



Experimental method

Same truth, same metrics, fair tuning

SGSMA 2026 · SANTIAGO DE CHILE



Slide 14 / 60 ~22s cum 5:45

CORE

Experimental method (1)

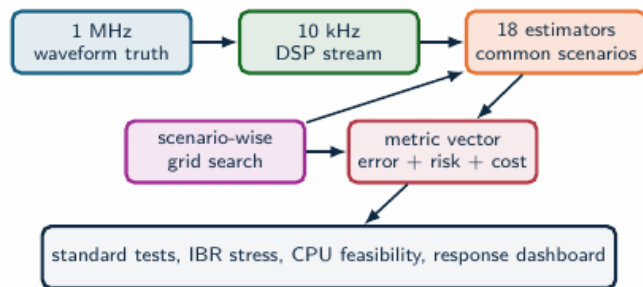
Say:

- Fairness is really the heart of it: every estimator sees the exact same input trace, the same random seeds, and the same warm-up handling.

→ *And we are just as strict about tuning and timing.*

Key line: Fairness is enforced by construction, not by trust.

Experimental method



Source: Methodology and test framework

Measurement setting

Local, non-time-synchronized relay-class frequency tracking from a decimated 10 kHz stream; PMU timestamp certification is outside scope.

Experimental principle

Each estimator–scenario pair is tuned within a declared grid, then compared using RMSE, peak error, trip exposure, settling, and CPU.

Slide 15 / 60 ~22s cum 6:07

Experimental method (2)

Say:

- We tune under a single fixed policy, we drop the warm-up region, and we record CPU cost as a genuine output, not an afterthought.

→ *Put it all together and this is the pipeline.*

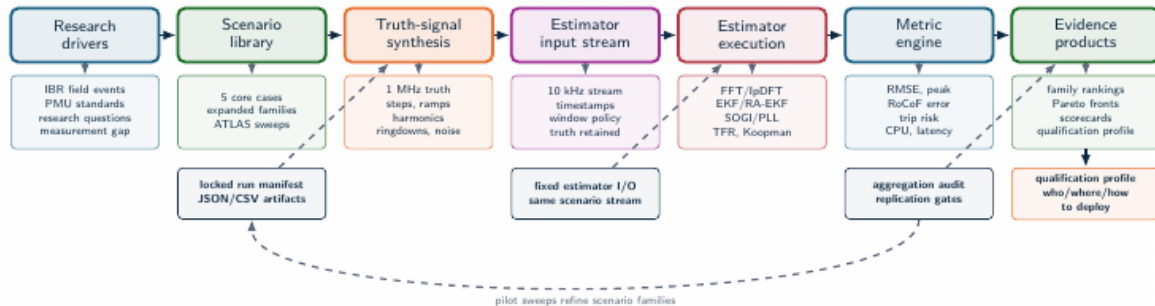
Key line: Nothing is hand-tuned to flatter one method.

Full benchmark workflow

Say:

- Scenario, Monte-Carlo, estimator, locked metrics, and then artifacts.
- Every single figure you will see today traces straight back to those artifacts.

→ *Let me introduce the scenarios themselves.*



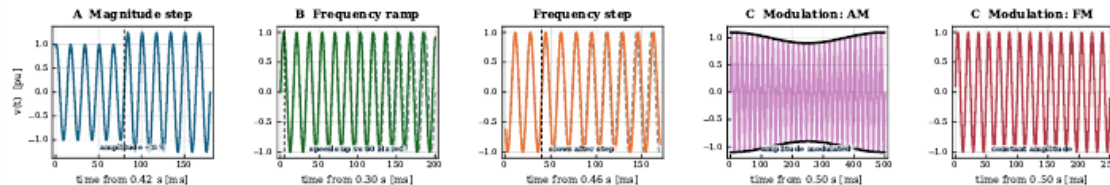
The benchmark is an evidence pipeline, not just a test script: every figure traces back to a locked manifest, a fixed I/O contract, and the readiness gate.

Key line: One reproducible pipeline, end to end.

Standard stresses: the waveform is controlled

CORE SCENARIO SUITE

Scenario suite I — standard stresses: voltage $v(t)$ (top) vs. true frequency $f(t)$ (bottom)



Each standard test changes one physical ingredient at a time: magnitude, ramp, frequency step, AM, or FM.

Standard stresses: controlled waveform

Say:

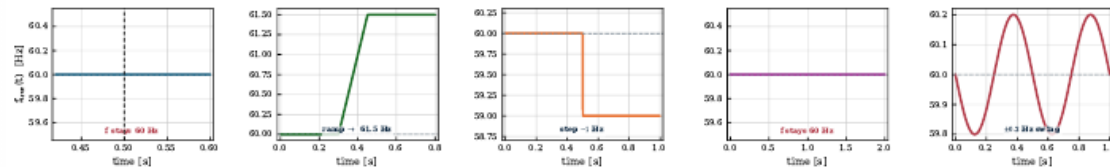
- We begin with the standard stresses: a step, a ramp, and modulation.
- Because the waveform is fully controlled, we know the true frequency at every single sample, so the error is exact.

→ Here is what they look like.

Key line: Known truth is what lets us score the error with no ambiguity.

Standard stresses: the frequency truth is simple

CORE SCENARIO SUITE



The clean reference traces make estimator failure visible: divergence is an algorithmic response, not ambiguity in ground truth.

Slide 18 / 60 ~30s cum 7:24

Scenario suite I (dashboard)

Say:

- Voltage on top, true frequency below, one column per scenario.
- And notice this: for the magnitude step and the AM case the frequency stays flat at sixty hertz, and yet the estimators still react to it.

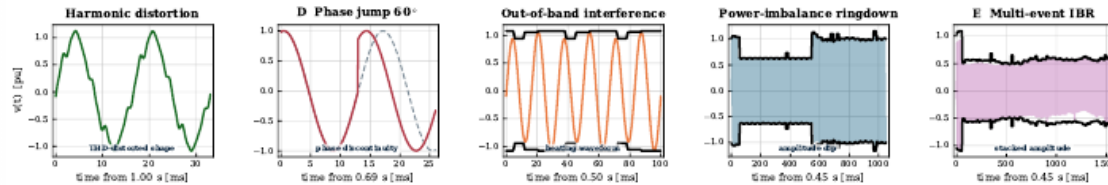
→ Now let us make it realistic.

Key line: The stress is not always a frequency event, but estimators react anyway.

Composite IBR stresses: waveform physics are stacked

IBR STRESS SUITE

Scenario suite II — composite IBR stresses: voltage $v(t)$ (top) vs. true frequency $f(t)$ (bottom)



Composite cases combine harmonic distortion, phase jumps, out-of-band tones, ringdown recovery, and multi-event amplitude dynamics.

Composite IBR stresses

Say:

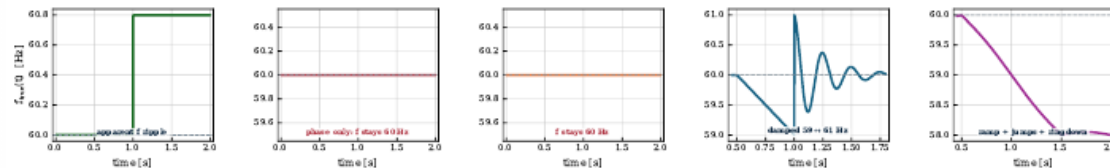
- Now we stack everything together: harmonics, phase jumps, a Ro-CoF segment, and ringdown, all in one waveform.
- This is much closer to what a real inverter-dominated fault actually produces.

→ Here is that composite suite.

Key line: Composite stress is the realistic, and the revealing, case.

Composite IBR stresses: frequency truth becomes diagnostic

IBR STRESS SUITE



The estimator must recover through nonstationary segments, not just report low steady-state error after the event.

- Same layout, voltage on top and frequency below, but these cases are deliberately harder than anything in a compliance test.

→ Before the results, a quick word on the methods and the metrics.

Key line: If a method survives here, it has earned our trust.

Estimator families

Loop-based

PLL, SOGI-PLL, SOGI-FLL, Type-3 SOGI-PLL.

Model-based / stochastic

EKF, UKF, RA-EKF, LKF, LKF2.

Legacy baselines

TKEO and ZCD are retained as comparison references outside the recommended deployment set.

Window / spectral

IpDFT, TFT, ESPRIT, Prony, MUSIC.

Adaptive and data-driven

RLS, Koopman (RK-DPMU).

Interpretation rule

These families occupy different latency and stress-sensitivity positions. The comparison tracks where each family wins, degrades, and fails.

LKF and LKF2 follow H. Ahmed, S. Biricik, M. Benbouzid, "Linear Kalman Filter-Based Grid Synchronization Technique: An Alternative Implementation," IEEE, 2021.

Estimator families

Say:

- The eighteen estimators sort into families: loop-based PLLs, window and spectral methods, model-based Kalman filters, adaptive methods, and data-driven ones.
- We analyze by family, because the family is what really explains the behavior.

→ And these are the numbers we score them on.

Key line: Eighteen estimators, five families, one fair contract.

Metrics and decision variables

Error and recovery

$$e[k] = \hat{f}[k] - f_{\text{true}}[k]$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_k e[k]^2}$$

$$E_{\text{max}} = \max_k |e[k]|$$

Protection-oriented risk

$$\tau_{\text{trip}} = \Delta t \sum_k \mathbf{1}\{|e[k]| > 0.5 \text{ Hz}\}$$

$$t_s = \min\{t : |e[\tau]| \leq \epsilon, \forall \tau \geq t\}$$

$$C_{\text{CPU}} = \text{mean time/sample}$$

The 0.5 Hz threshold is a protection-oriented risk proxy. The metric set separates average error, transient severity, trip exposure, settling, and compute cost.

Metrics and decision variables

Say:

- We track RMSE for average error, peak error, and a trip-risk time that counts how long the error stays above half a hertz.
- And then settling time and CPU cost.

→ Now, let me show you what the data actually says.

Key line: Half a hertz is our protection-oriented risk proxy.

Core evidence (section)

Key line: The core evidence: five scenarios, from standard to composite.

Core evidence

From standard cases to composite IBR stress

Standard scenario results

Method	Step	Ramp	Mod.
RA-EKF	0.000	0.006	0.037
SOGI-FLL	0.005	0.013	0.040
Koopman	0.003	0.017	0.044
TFT	0.001	0.019	0.046
UKF	0.000	0.017	0.055
LKF2	0.138	0.015	0.028
PLL	0.033	1.47	0.227
IpDFT	0.000	14.4	0.114
EKF	0.000	37.1	0.064

RMSE [Hz], mean over 5 seeds. Green:

below 0.05 Hz guide; red: failure/divergence regime.

What the heatmap shows

The standard step is easy for most methods. The ramp is the separator: EKF and IpDFT can look excellent at a point and still diverge in another seed.

Operational consequence

Robustness is not the same as point accuracy. A standard scenario can pass while the same estimator remains unsafe for ramp-like IBR dynamics.

Result: RA-EKF, SOGI-FLL, Koopman, and TFT remain below 0.05 Hz across the three standard cases; EKF and IpDFT fail through intermittent ramp divergence.

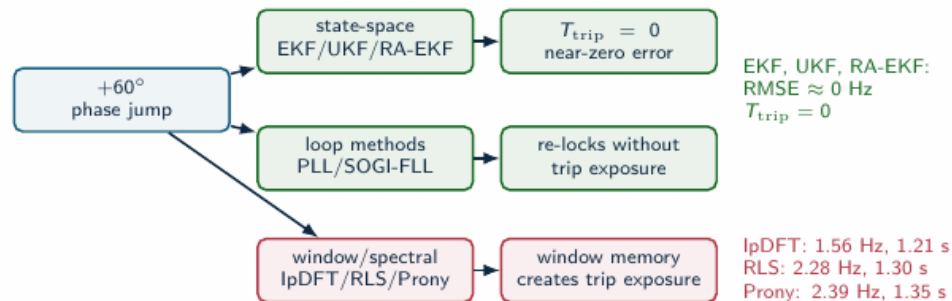
Say:

- On the standard cases, almost everything looks excellent, all below fifty millihertz.
- CLICK. But now watch the ramp: the textbook EKF and IpDFT intermittently blow up, to thirty-seven and fourteen hertz.

→ *You would expect a phase jump to be even worse, so let me show you.*

Key line: Robustness, not point accuracy, is what separates the families.

A clean phase jump is a family separator



The clean jump does not expose RA-EKF's main advantage. It shows which estimator families carry memory through discontinuities.

Scenario D: phase jump separates families

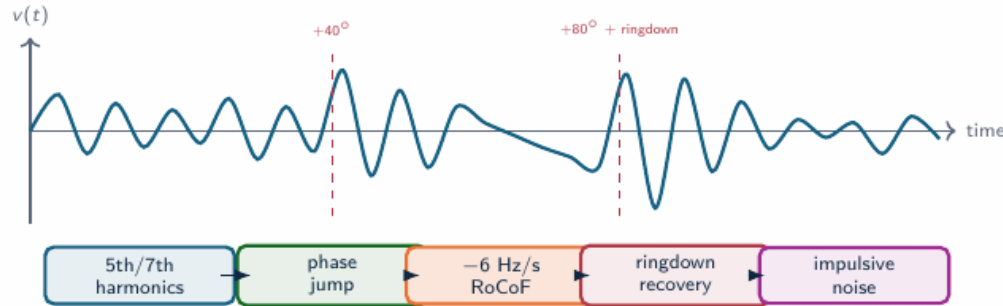
Say:

- A clean sixty-degree phase jump splits the families cleanly.
- The state-space filters and the PLLs absorb it with zero trip exposure.
- The windowed and spectral methods, though, carry over a second of exposure.

→ But honestly, a single jump is not the real challenge.

Key line: Here is the surprise: the phase jump does not break the EKF.

Composite IBR sequence: why it is hardest



Why standard tests miss it

The estimator sees spectral leakage, loop re-lock, state-model mismatch, transient recovery, and noise in one nonstationary waveform.

Why ATLAS decomposes it

The multi-event case proves the failure exists; severity sweeps identify which stressor actually breaks each estimator family.

Slide 26 / 60 ~25s cum 10:56

CORE

Composite IBR sequence: hardest case

Say:

- Scenario E stacks it all: harmonics, two phase jumps, a RoCoF segment, ringdown, and impulsive noise.
- So spectral leakage, loop re-lock, and model mismatch all hit at the very same time.

→ And this is where the leaderboard falls apart.

Key line: This is the hardest, most realistic case in the whole suite.



Scenario E: the leaderboard breaks

Method	RMSE [Hz]	T_{trip} [s]	CPU [μ s]
UKF	1.09	0.78	14
EKF	1.13	0.87	11
SOGI-PLL	1.14	0.78	11
SOGI-FLL	1.14	0.84	9
lpDFT	1.14	1.35	17
RA-EKF	1.16	0.92	15
Koopman	1.90	0.84	15 241
ZCD	426	1.55	7

No accurate winner

Under the stacked sequence every usable method collapses to ≈ 1.1 Hz RMSE; ZCD diverges catastrophically (426 Hz). High accuracy is simply not available here.

Source: full_mc_benchmark, $n = 5$: IBR_Multi_Event

Scalar ranking failure

UKF has the lowest RMSE and trip exposure, SOGI-FLL the lowest usable CPU, Koopman is 1000 $\times$ costlier; RA-EKF sits mid-pack but never diverges.

Scenario E: the leaderboard breaks

Say:

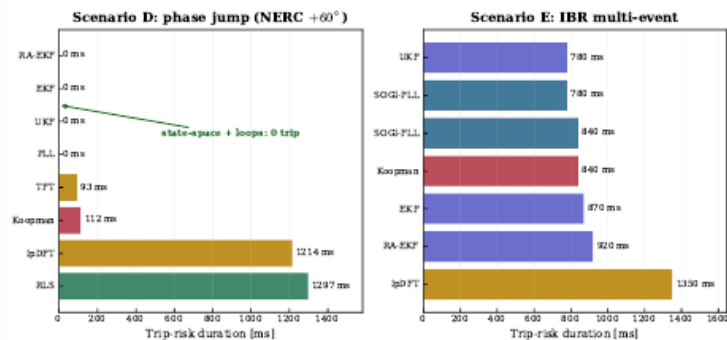
- Every usable method collapses to about one-point-one hertz, and ZCD diverges all the way to four hundred.
- Worse, the rankings by RMSE, by trip-risk, and by CPU now disagree with each other.

→ Let me make the trip-risk part of that visual.

Key line: Under stacked stress, there is simply no accurate winner.

Trip-risk evidence: core stress cases

Trip-risk is an event metric, not an average-error metric



Result reading

- Scenario D: state-space filters and PLL keep $T_{\text{trip}} = 0$; windowed/spectral methods (lpDFT, RLS) exceed 1.2 s.
- Scenario E: every method carries 0.8–1.4 s exposure — composite stress is unavoidable.
- The trip-exposure ranking is not the RMSE ranking.

Result: T_{trip} separates protection exposure from average error.

Trip-risk evidence (figure)

Say:

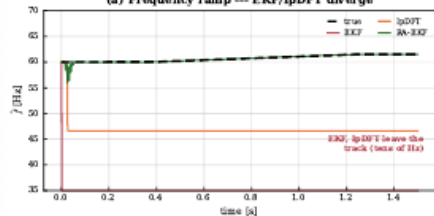
- In Scenario D the state-space filters hold zero exposure while the windowed methods sail past 1.2 seconds.
 - In Scenario E, though, everyone carries exposure.
- And you can actually watch the divergence happen.

Key line: Trip-risk is a completely different ranking from RMSE.

The failures are visible in the trace, not only in the table

RAMP DIVERGENCE

Event responses (full_mc_benchmark): intermit

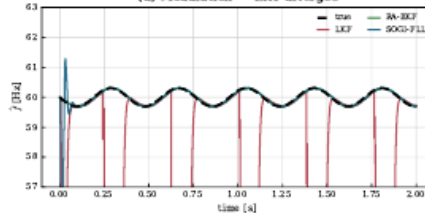


Failure signature

EKF and IpDFT leave the true ramp almost immediately; after the event, the estimate is no longer operationally usable.

MODULATION DIVERGENCE

(d) Modulation --- LKF diverges



Failure signature

LKF stays plausible between collapses, then repeatedly drops out of the usable frequency band.

Event responses (figure)

Say:

- True frequency in black, and each estimator trying to follow it.
- You can literally see EKF and IpDFT run away on the ramp, while RA-EKF just stays locked.

→ So what does all of this mean when you actually have to choose?

Key line: This is the intermittent divergence, with your own eyes.

Result comparison: winners depend on the question

Question	Best-supported answer	Evidence	Boundary
Clean step/modulation accuracy	State-space + SOGI-FLL near-zero; LKF2 best on mod	Step ≈ 0 ; mod 0.028–0.044 Hz	Clean cases do not predict ramp/event survival
Ramp and RoCoF tracking	RA-EKF is the most robust ramp tracker	0.006 Hz vs EKF 37 Hz, IpDFT 14 Hz (diverge)	Divergence is intermittent (seed-dependent)
Phase jump (Scenario D)	State-space filters recover with zero trip	EKF/UKF/RA-EKF $T_{\text{trip}} = 0$; IpDFT 1.21 s	Windowed/spectral carry the exposure
Multi-event RMSE	No accurate method — all ≈ 1.1 Hz	UKF 1.09 Hz best; ZCD 426 Hz diverges	RMSE hides trip-risk and 1000× CPU spread
Embedded feasibility	PLL, SOGI-FLL, EKF, RA-EKF remain plausible	9–17 $\mu\text{s}/\text{sample}$; Koopman $\sim 15,000$	Python timing is a software-cost proxy

Selection changes with event family, objective metric, and compute envelope. A single leaderboard hides the deployment trade-off.

Result comparison: winners by question

Say:

- Ask for clean accuracy, and you get the state-space methods and SOGI-FLL.
- Ask for ramp robustness, and it is RA-EKF. Ask about multi-event, and there is no winner. Ask about feasibility, and the expensive methods simply drop out.

→ *And feasibility brings us straight to cost.*

Key line: Different question, different winner, every single time.

Computational feasibility

Method	Time/sample	Family	Target
ZCD	7 μ s	Loop	MCU
SOGI-FLL	9 μ s	Loop	MCU
PLL	9 μ s	Loop	MCU
EKF	11 μ s	Model	MCU/DSP
UKF	13 μ s	Model	DSP
RA-EKF	15 μ s	Model	DSP
IpDFT	17 μ s	Window	DSP/FPGA
ESPRIT	$\approx$ 1,400 μ s	Window	CPU
Koopman	$\approx$ 15,000 μ s	Data-driven	CPU/GPU

Source: full_mc_benchmark, $n = 5$: mean m13_cpu_time_us

Deployment interpretation

The Python baseline separates feasible estimator families from clearly prohibitive ones; hardware latency requires platform timing.

Key comparison

The state-space and loop families all run at $\sim$ 10–17 μ s/sample (embedded-feasible); spectral (ESPRIT, MUSIC) and data-driven (Koopman) methods are 100–1000 $\times$ costlier and data-dependent.

Computational feasibility

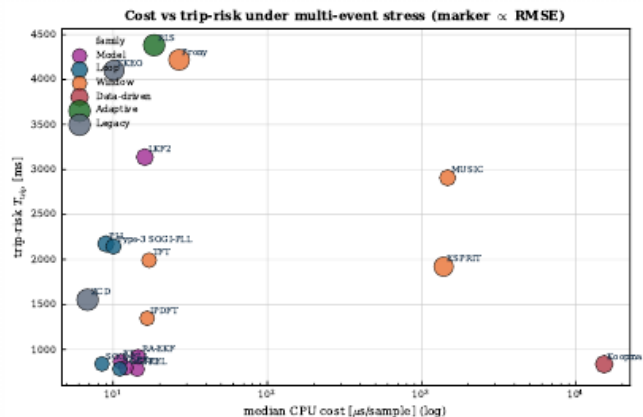
Say:

- The loop and state-space methods run in nine to seventeen microseconds per sample, which is embedded-feasible.
- The spectral and data-driven methods are a hundred to a thousand times slower than that.

→ Put cost and risk on the same picture and it gets clearer still.

Key line: Cost alone already rules several methods out of a relay.

Deployment limits



Deployment interpretation

- Deployment combines RMSE, trip exposure, CPU, and divergence robustness.
- State-space + loop methods cluster at low cost ($\leq 17 \mu s$) and low trip.
- Spectral/data-driven methods trade lower error for 100–1000 $\times$ higher CPU.
- RA-EKF stays low-cost with no ramp divergence in this run.

Slide 32 / 60 ~25s cum 13:54

FIG

Deployment limits (figure)

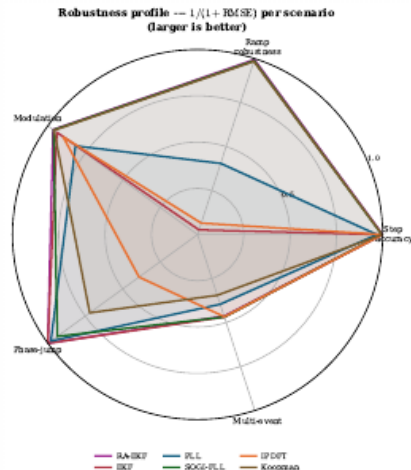
Say:

- Deployment is really the combination of accuracy, trip exposure, cost, and divergence-robustness.
- And notice that the low-cost cluster, the state-space and loop methods, is also the low-trip cluster.

→ We can capture each method's balance in a single shape.

Key line: The practical winners sit together, at low cost and low trip.

Balance profile



Radar profile

- Each axis is one scenario; larger = more robust ($1/(1 + RMSE)$).
- RA-EKF and SOGI-FLL stay robust across all five scenarios.
- EKF and IpDFT collapse on the ramp axis (intermittent divergence).
- No estimator is strong on every axis — robustness is the discriminator.

Slide 33 / 60 ~25s cum 14:19

Balance profile (radar)

Say:

- Each axis is one scenario, and bigger means more robust.
- RA-EKF and SOGI-FLL stay robust on every axis, while EKF and IpDFT collapse on the ramp axis.

→ Now, this next slide is the most important one about compliance testing.

Key line: No method is strong on every axis.

Compliance heatmap: why standard tests are insufficient

Method	Step	Ramp	Mod	Jump	Multi
Koopman	P	P	P	F	F
TFT	P	P	P	F*	F
IpDFT	P	F	P	F	F
SOGI-FLL	P	P	P	P	F
PLL	P	F	F*	P	F
EKF	P	F	P	P	F
UKF	P	P	P	P	F
RA-EKF	P	P	P	P	F
LKF2	F*	P	P	P	F

Columns: **Jump** = phase jump (D), **Multi** = multi-event (E). **P** pass | **F** fails a declared RMSE/ T_{trip} /settling threshold | **F*** borderline.

Result: Step, ramp, and modulation compliance does not predict phase-jump or multi-event readiness.

Compliance heatmap (build-up)

Say:

- Look at the standard columns first: nearly everyone passes step, ramp, and modulation.
- CLICK. But watch, the same methods now fail on phase jump and on multi-event.

→ Let me distill that into our empirical claims.

Key line: Passing the standard tests does not predict event readiness.



Empirical claims

Protection behavior

Divergence, not phase jumps, is the dominant failure mode.
On the ramp, EKF and IpDFT intermittently diverge (37 and 14 Hz mean); RA-EKF stays at 0.006 Hz. A clean phase jump is recovered by every state-space filter with $T_{\text{trip}} = 0$.

Standard-test limitation

Simple-case pass/fail leaves event readiness unresolved.
The heatmap shows methods that pass Step/Ramp/Mod but fail islanding or multi-event IBR stress.

Guardrail: report scenario, metric, and compute envelope before naming a preferred estimator.

Metric disagreement

RMSE, trip-risk, and CPU select different winners.
Under multi-event stress UKF has the lowest RMSE/trip, SOGI-FLL the lowest usable CPU, and Koopman is 1000× costlier.

Deployability

Feasibility changes the accuracy ranking.
RA-EKF runs at $\sim 15 \mu\text{s}/\text{sample}$; the spectral and data-driven methods are 100–1000× costlier and sit outside relay-class timing.

Say:

- Four claims. First, divergence, not phase jumps, is the dominant failure mode.
- And then: pass-fail leaves readiness unresolved, the three metrics pick different winners, and feasibility changes the ranking.

→ *Now the natural question: does this pattern survive when we scale up?*

Key line: Four claims, one theme: always report the conditions.

Expanded and ATLAS (section)

Expanded & ATLAS

Does the pattern scale across families and severity?

Key line: Does the pattern hold across families and severity? That is ATLAS.



Expanded stress analysis

FULL MONTE-CARLO BENCHMARK: 18 estimators, 33 scenarios, $n = 5$ per scenario

Monte-Carlo benchmark (full_mc)

- 18 estimators across 5 families.
- 33 scenarios (A–E plus severity variants).
- $n = 5$ Monte Carlo seeds per scenario.
- Standard, phase-jump, multi-event, and CPU metrics.

ATLAS severity sweeps

- Severity, sign, and policy curves per stress family.
- Fixed-policy, paper-grade $n = 30$ across all 9 required sweeps (both signs).
- Harmonic, RoCoF, phase-jump, and noise atlases.
- Tests sensitivity, scaling, and family stability.

Two evidence layers on the same truth: a replicated Monte-Carlo benchmark, plus ATLAS severity sweeps that trace where each estimator family degrades.

Slide 37 / 60 ~22s cum 15:52

CORE

Expanded stress analysis

Say:

- Separate the evidence layers: the SGSMA benchmark matrix is sixteen estimators, thirty-four scenarios, and sixty Monte Carlo runs; the OpenFreqBench expansion is eighteen estimators, thirty-three scenarios, five seeds.
- ATLAS then adds severity and sign sweeps on top of those benchmark results.

→ *First question: do the family winners stay fixed?*

Key line: We scale the test, we do not just repeat it.

Expanded benchmark: family winners

FULL MONTE-CARLO BENCHMARK: 18 estimators, 33 scenarios, $n = 5$ per scenario

Family	Count	RMSE leader	RMSE	Trip-risk leader
IEEE ramps	9	RA-EKF	11 mHz	0 s (state-space)
Magnitude steps	7	UKF	<1 mHz	0 s (state-space)
Modulation	3	LKF2	15 mHz	0 s (state-space)
Phase jumps	3	UKF	<1 mHz	0 s (state-space)
Frequency step	1	RA-EKF	52 mHz	RA-EKF, 3 ms
OOB interference	1	UKF	<1 mHz	0 s (state-space)
IBR harmonics	3	SOGI-FLL	61 mHz	SOGI-PLL, 10 ms
IBR ringdown	4	LKF2	202 mHz	ESPRIT, 87 ms
IBR multi-event	1	UKF	1.09 Hz	UKF, 780 ms

The full Monte-Carlo sweep confirms the central pattern: the RMSE leader rotates (RA-EKF, UKF, LKF2, SOGI-FLL) by disturbance family, and error grows from sub-mHz on clean cases to ~ 1 Hz under multi-event stress.

Expanded benchmark: family winners

Say:

- They do not stay fixed. The winner rotates: one method leads the step, another the ramp, another harmonics, another the multi-event.

→ *ATLAS lets us see why, by sweeping the severity.*

Key line: The leaderboard is regime-dependent, never absolute.

ATLAS severity analysis

Say:

- ATLAS sweeps each stress by severity and by sign, under a fixed policy, at thirty Monte-Carlo runs.
- And this is now validated across all nine required sweeps.

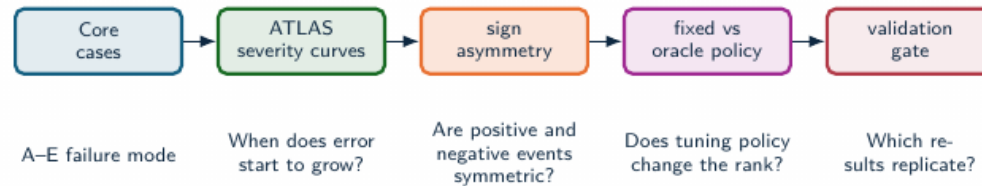
→ *And the headline result here is genuinely striking.*

Key line: This is replicated severity evidence, not a pilot.

ATLAS severity analysis



PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)



ATLAS maps where estimator behavior changes with severity, sign, tuning policy, and replication depth.

Source: docs/ATLAS_PIPELINE.md; docs/ATLAS_METHOD_AUDIT.md



ATLAS result: metric choice changes the answer

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

Comparison	Different winner	Levels tested
RMSE vs trip-risk	83	113
RMSE vs RoCoF max error	55	113
RMSE vs RoCoF RMS error	50	113

Winner concentration

RMSE is led mostly by EKF and ESPRIT. Trip-risk is led mostly by Koopman, EKF, SOGI-FLL, ZCD, and ESPRIT. CPU is always led by ZCD.

Result: across 113 severity levels, the preferred estimator changes when the objective changes.

Source: atlas-papergrade-missing-v1/global_metrics_report.csv

Why this matters

For protection-adjacent use, a low average error is not enough. The estimator must avoid long deadband exposure, track RoCoF, and remain computable.

Operational implication

The benchmark output is not a leaderboard. It is a metric-conditioned selection map.

ATLAS: metric choice changes the answer

Say:

- Across a hundred and thirteen severity levels, the preferred estimator changes the moment the objective changes.
- Trip-risk has many tied zero-exposure minima, so I read it as an exposure screen. RoCoF is the stronger rank-changing metric: it differs from the RMSE leader in roughly half the severity map.

→ *ATLAS also tells us where each method gives up.*

Key line: The output is a selection map, not a leaderboard.



ATLAS result: where estimators stop being valid

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

Stress family	Maximum tested level	Methods below 0.05 Hz	Interpretation
Magnitude step	$\pm 90\%$ sag/swell	2	EKF and RA-EKF survive both signs.
AM modulation	10 Hz envelope	10	AM alone is not the hardest modulation case.
FM modulation	10 Hz modulation	0	FM is qualitatively harder than AM.
RoCoF ramp	± 50 Hz/s	0	Severe RoCoF requires a separate qualification class.
Phase jump	$\pm 60^\circ$	0	Single jump severity already breaks the 0.05 Hz guide.
Harmonics	30% THD	0	Low-THD winners do not survive extreme THD.
Interharmonics	20% at 75 Hz	0	Integer-harmonic tests are not sufficient.
Broadband noise	$\sigma = 0.1$ pu	0	Noise robustness is a first-class stress dimension.

ATLAS: where estimators stop being valid

Say:

- For every method, ATLAS pinpoints the severity at which it crosses over into failure.

→ *Let me show you a few of the most surprising sweeps.*

Key line: That validity boundary is different for every family.

AM, FM, and interharmonics split the story

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

Stress	Winner pattern	Threshold behavior
AM modulation	EKF wins slow, reference, fast, and severe regions.	Ten methods stay below 0.05 Hz through 10 Hz envelope modulation.
FM modulation	TFT leads slow; ESPRIT leads reference/fast/severe; SOGI-PLL reaches the best threshold.	No method stays below 0.05 Hz at 10 Hz FM modulation.
Interharmonics	EKF wins trace, reference, high, and severe levels.	EKF crosses the 0.05 Hz guide around 8%; most other families cross earlier.

Implication

AM, FM, integer harmonics, and interharmonics should not be collapsed into one “distortion” label. They select different estimator families.

Decomposition insight

A composite IBR trace can contain AM, FM, harmonics, interharmonics, noise, and phase steps. ATLAS isolates which stressor changes the ranking.

AM, FM and interharmonics

Say:

- Even modulation splits the story: AM is easy, and there the model-based methods win.
- FM and interharmonics are harder, and there the spectral methods take over.

→ Now the handful of findings I most want you to remember.

Key line: Even inside modulation, the winner switches.

When the textbook filter diverges

FULL MONTE-CARLO BENCHMARK: 18 estimators, 33 scenarios, $n = 5$ per scenario

Per-seed frequency RMSE [Hz] on the IEEE frequency-ramp scenario (5 Monte Carlo seeds):

Estimator	seed 1	seed 2	seed 3	seed 4	seed 5	trip-risk
EKF (textbook)	0.016	63.7	0.017	121.5	0.027	up to 1.35 s
RA-EKF	0.0055	0.0062	0.0083	0.0045	0.0058	0 s

What happens

The standard EKF tracks most seeds but **intermittently diverges** to tens of Hz. It is *not* a severity threshold: the worst seed (121 Hz) had a low ramp rate (1.7 Hz/s) while the steepest (5 Hz/s) tracked cleanly. The failure is triggered by onset/phase combinations — seed-dependent and silent.

Why RA-EKF matters here

Covariance scaling and event gating remove the divergence: stable on **every** seed, zero trip exposure. This — not a universal RMSE win — is its defensible value.

When the textbook filter diverges

Say:

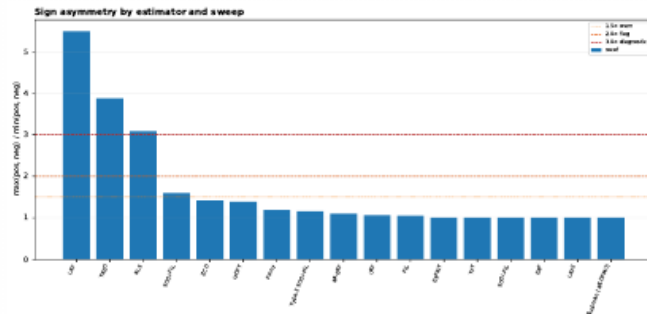
- The textbook EKF tracks most seeds, and then intermittently runs away to tens of hertz.
- And it is not severity-driven, because the worst seed was actually the one with the gentlest ramp.

→ *The next finding is all about direction.*

Key line: The robustified RA-EKF removes the divergence on every seed.

RoCoF has a sign: down-ramps are harder

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)



Source: atlas-papergrade-missing-v1/atlas_sign_asymmetry.csv ($n = 30$, both signs)

Directional bias (RMSE₋/RMSE₊ at ± 50 Hz/s)

- LKF $5.5\times$ worse on $-RoCoF$
- TKEO $3.9\times$, RLS $3.1\times$
- SOGI-FLL $1.6\times$
- EKF, UKF, RA-EKF ≈ 1.0 (symmetric)

Why it matters

A $+RoCoF$ -only test can certify a method that fails on frequency *decline* — the protection-critical direction. Both signs must be swept.

RoCoF has a sign

Say:

- RoCoF has a sign, and it turns out down-ramps are harder than up-ramps.
- LKF is five-and-a-half times worse when frequency falls; TKEO almost four times worse.

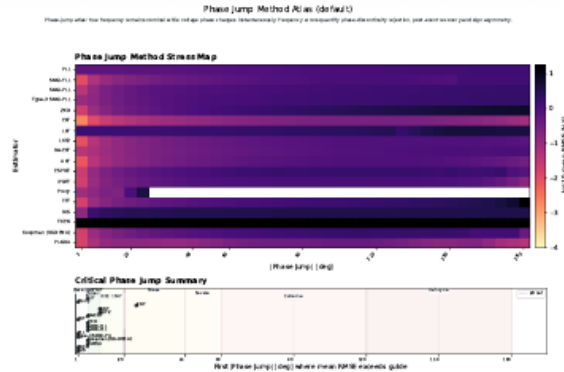
→ And severity can surprise us in the opposite direction too.

Key line: A rising-only test can certify a method that fails when frequency drops, the protection-critical case.



Phase-jump severity is non-monotone

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



Worst case is not 180°

Most methods peak near 90–120° and *recover* toward 180° (a half-cycle jump partly self-cancels). EKF peaks at only 0.25 Hz @ 115°, back to 0.07 Hz @ 180°.

The TFT cliff

One exception: TFT rises gently, then jumps 2.5 → 21.9 Hz between 150° and 180° — a phase-reversal singularity.

Result: Severity is non-monotone; sweep the jump angle rather than qualify one nominal point.

Say:

- Phase-jump severity is non-monotone: the worst case is not a hundred and eighty degrees.
 - Most methods peak around ninety to a hundred and twenty degrees and then recover, because a half-cycle jump partly cancels itself out.
- *Severity also reshuffles the winners completely.*

Key line: Bigger is not always worse, so you have to sweep the range.

No estimator owns the whole stress space

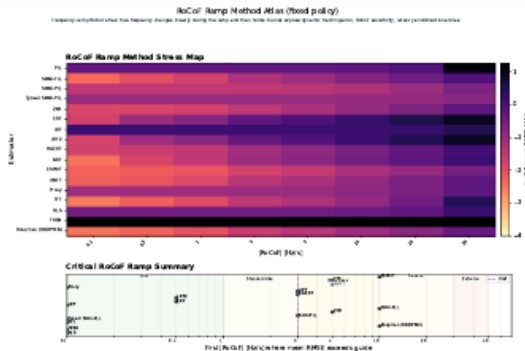
PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

RoCoF regime	Best	med. RMSE
Low (0.1–1 Hz/s)	Koopman	4.1 mHz
Standard (1–3)	ESPRIT	7.6 mHz
IBR stress (3–10)	ESPRIT	22.3 mHz
Severe (10–30)	SOGI-PLL	70.4 mHz

Across stress types

EKF wins phase jumps, the magnitude step and interharmonics; ESPRIT/Koopman win the RoCoF and frequency-step regimes; ZCD wins low-THD harmonics; SOGI-PLL leads severe RoCoF. No method leads everywhere.

The deliverable is a selection map: event class $\times$ metric $\times$ compute budget.



No universal winner (map)

Say:

- This map really says it all: the champion rotates by RoCoF regime.
- Koopman at low rates, ESPRIT in the middle, SOGI-PLL once it gets severe.

→ And one estimator manages to be both the best and the worst.

Key line: No universal winner; the deliverable is the map itself.

ZCD is a narrow instrument, not a general estimator

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

Low-stress champion

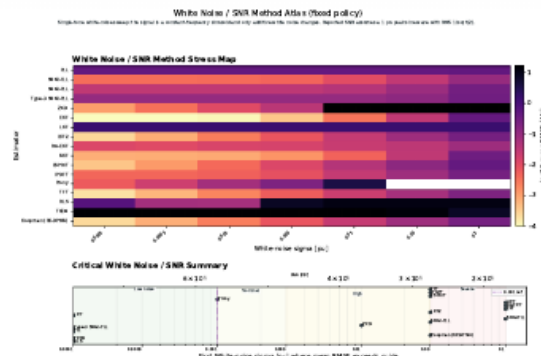
Up to $\sim 15\%$ THD, zero-crossing detection gives the *lowest* error of any method — **0.27 mHz** at 1% THD, 2.2 mHz at 15%; and just 1.1 mHz at low broadband noise.

Broadband-noise cliff

Same method: 34 mHz at $\sigma=0.003$ pu $\rightarrow$ **3112 Hz** at $\sigma=0.1$ pu. EKF stays < 0.43 Hz across the same sweep.

Harmonic cliff

Same method again: 2.2 mHz at 15% THD → 39 Hz at 20% → 86 Hz at 30%.



Say:

- Below fifteen percent harmonic distortion, zero-crossing detection is the single most accurate method we have.
- CLICK. Add broadband noise, and the same method explodes to three thousand hertz.
- CLICK. Push past twenty percent distortion, and it collapses to tens of hertz.

→ And that same asymmetry shows up in voltage events.

Key line: The very same estimator is the best and the worst, depending on the stress.



Voltage sags and swells are not symmetric

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

Estimator at $\pm 90\%$	Sag to 0.1 pu	Swell to 1.9 pu
EKF RMSE	47 mHz	14 mHz
RA-EKF RMSE	49 mHz	30 mHz
ZCD RMSE	374 Hz	9.4 mHz
ZCD peak error	2856 Hz	32 mHz

Mechanism

Zero-cross timing error scales roughly as $\sigma_n/(A\omega)$. A deep sag lowers zero-cross slope and makes false crossings likely when absolute noise is fixed.

Source: atlas-papergrade-nissing-v1/global_metrics_report.csv; src/estimators/zcd.py

What to claim

ZCD is amplitude-invariant only in the noise-free ideal case. Under fixed absolute noise, deep voltage sags create a low-SNR crossing problem.

Benchmark consequence

Sag and swell signs must be swept separately; a positive-only magnitude-step test can hide the worst case.

Slide 48 / 60 ~25s cum 20:57

CORE

Sags and swells are not symmetric

Say:

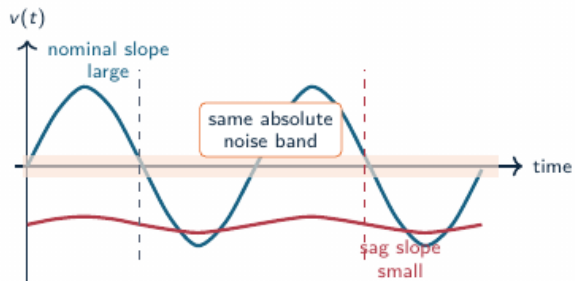
- Voltage magnitude is not symmetric either: a deep sag is far harder than a swell.
- ZCD goes from millihertz on a swell to hundreds of hertz on a deep sag.

→ And there is a clean physical reason for that.

Key line: Direction matters, so you have to sweep both signs.

Why a voltage sag breaks zero-crossing detection

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)



Timing sensitivity

$$\Delta t \approx \frac{n(t_0)}{\left. \frac{dv}{dt} \right|_{t_0}} \quad \left. \frac{dv}{dt} \right|_0 \approx A\omega$$

When the voltage amplitude A collapses, the same noise produces much larger crossing-time error.

Estimator consequence

ZCD is elegant at clean high-SNR crossings, but deep sags convert amplitude loss into apparent frequency jitter and false crossings.

Why a sag breaks zero-crossing

Say:

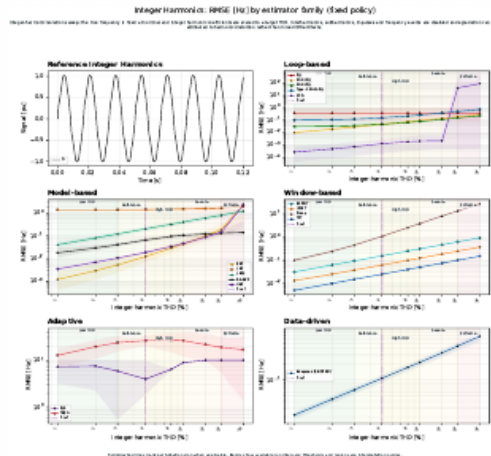
- The timing error of a zero-crossing scales with noise over amplitude times frequency.
- A deep sag flattens the slope right at the crossing, so the noise starts creating false crossings.

→ *Harmonics tell a similar, family-by-family story.*

Key line: This is physics, not a coding bug.

Harmonic degradation differs by estimator family

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)



Result

- Loop, window, model, and adaptive families degrade with THD at different rates.
- Best low-THD accuracy does not predict high-THD survival.
- Loop methods (ZCD) win at low THD but collapse above $\sim 15\%$; window methods (TFT) degrade gently and lead at extreme THD.

Slide 50 / 60 ~25s cum 21:47

CORE

Harmonic degradation by family

Say:

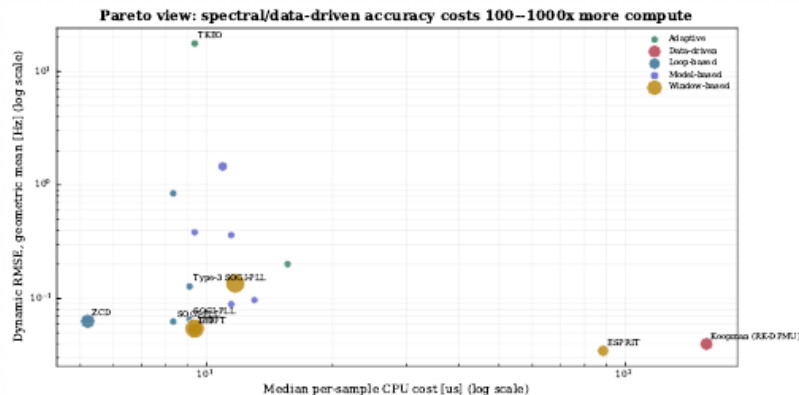
- Harmonic degradation depends on the family: loop methods like ZCD win at low distortion but collapse above fifteen percent.
- Window methods degrade gently and end up leading at extreme distortion.

→ And cost is the final dimension.

Key line: Low-distortion accuracy does not predict high-distortion survival.

Accuracy costs 100–1000× more compute

FULL MONTE-CARLO BENCHMARK: 18 estimators, 33 scenarios, $n = 5$ per scenario



Low-cost core
PLL, SOGI-FLL, EKF, and RA-EKF sit in the deployable compute range.

Accuracy premium
Spectral and data-driven methods can move right by 100–1000× CPU cost.

CPU-accuracy Pareto

Say:

- On the cost-accuracy Pareto, the spectral and data-driven methods buy you a little accuracy for a hundred to a thousand times more compute.

→ So, with all of that, what do we actually deploy?

Key line: For relay-class timing, that trade is usually not worth it.



Findings & deployment

What to deploy — and what comes next

SGSMA 2026 · SANTIAGO DE CHILE



Slide 52 / 60 ~6s cum 22:15

QUICK

Findings and deployment (section)

Key line: So what do we actually deploy?



Three findings

1. Metric-conditioned

The RMSE leader often differs from the trip-risk and RoCoF leader across ATLAS severity levels.

2. Stress-bounded

AM remains survivable for many methods; severe FM, RoCoF, interharmonics, noise, and extreme THD break the 0.05 Hz guide.

3. Robustness over point accuracy

EKF, UKF, and IpDFT can diverge on ramps; RA-EKF keeps the ramp family stable across seeds.

Dynamic frequency estimators should be qualified by event family, sign, severity, recovery, trip exposure, and compute envelope.

Say:

- Three findings. First, there is a clear latency-versus-stress trade-off.
- Second, robustness beats point accuracy: RA-EKF's explicit RoCoF state keeps it stable exactly where EKF and IpDFT diverge.
- And third, compliance tests are simply not enough for inverter events.

→ *Let me be honest about the limits.*

Key line: Robustness, not accuracy, is the deployment discriminator.



Validity limits and next steps

Validation limits

- Limited to the reported scenario set.
- CPU is a software-level cost indicator; hardware latency requires platform timing.
- The study is simulation-only.
- Scenario-wise tuning can affect rank stability.

Validation program

- Hardware-in-the-loop validation.
- Wider converter-control interaction coverage.
- Sensitivity analysis of the tuning protocol.
- ATLAS severity and sign sweeps with validation gates.

These limits define the validation path: broader event coverage, hardware timing, fixed-policy replication, and field replay.

Slide 54 / 60 ~28s cum 23:18

CORE

Validity limits and next steps

Say:

- On validity: the SGSMA benchmark matrix is sixty runs, the Open-FreqBench expansion is five seeds, and the ATLAS sweeps are validated at thirty.
- The next step is an $n \geq 100$ replication, plus the subspace and machine-learning methods.

→ *And from there it becomes a concrete recommendation.*

Key line: We are explicit about what is diagnostic and what is replicated.

Recommendation map

Say:

- For fast frequency response and anti-islanding, RA-EKF is the strongest benchmark-backed choice.
- PLL and EKF make solid low-cost baselines, and IpDFT or TFT work where some observation delay is acceptable.

→ And we wrap that into a qualification idea.

Key line: We deliver a recommendation map, not a single ranking.

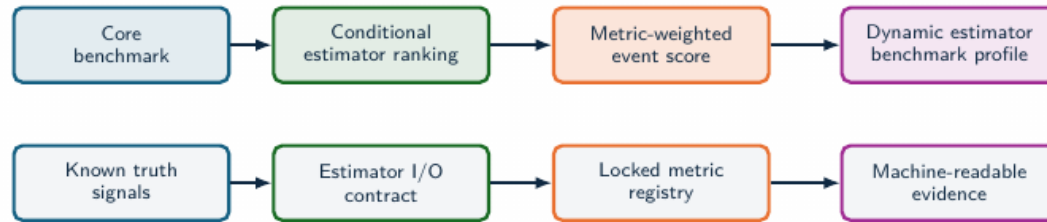
Recommendation map



Deployability depends on event class, trip-risk tolerance, and compute budget.



Qualification profile



Estimator selection moves toward **qualification** when each method is tested against a declared event profile, metric profile, and latency-cost envelope.

Say:

- A real qualification profile should test transient recovery, trip exposure, compute feasibility, and standard-case RMSE, together.

→ *Reported in a standard, comparable form.*

Key line: Qualify on all dimensions at once, not one at a time.

Standardization profile



Current standardization layer

- PMU standards define phasor, frequency, and RoCoF measurement requirements.
- Compliance tests are necessary, with estimator-family ranking by IBR event survival left outside scope.
- Device reports often hide algorithmic failure modes behind aggregate accuracy.

Standardization boundary

A dynamic-estimator profile needs declared signals, estimator I/O, metrics, evidence artifacts, and latency-cost reporting.

Benchmark profile

- Canonical events: steps, ramps, phase jumps, ringdown, harmonics, interharmonics, noise, and composite IBR sequences.
- Locked metrics: RMSE, peak error, RoCoF error, trip-risk, settling, invalid samples, CPU, and latency.
- Task profiles: forensic monitoring, PMU augmentation, DFR support, and relay-adjacent screening.

Standardization value

A reference test profile needs the same signals, estimator I/O, metrics, and machine-readable evidence.

Slide 57 / 60 ~20s cum 24:30

Standardization profile

Say:

- So we propose reporting these dimensions explicitly, so that any two estimators can be compared on equal terms.

→ And when you truly need just one number, here it is.

Key line: Make the comparison reproducible and fair.



IBR Event Qualification Score

COMPOSITE EVENT SCORE: weighted event-class qualification metric

Score for estimator e

$$\text{IBR-ERS}(e) = 1 - \sum_{s \in S} w_s \Phi_s(e),$$

$$\Phi_s(e) = \alpha \tilde{E}_s + \beta \tilde{P}_s + \gamma \tilde{T}_s + \delta \tilde{S}_s + \eta \tilde{C}_s + \lambda \tilde{L}_s.$$

Score composition

RMSE	peak	trip	settle	cost
$\tilde{E}_s, \tilde{P}_s$	dynamic and peak error			
$\tilde{T}_s, \tilde{S}_s$	trip exposure and recovery			
$\tilde{C}_s, \tilde{L}_s$	compute and latency penalties			

Why a score is needed

RMSE, trip-risk, transient recovery, and cost lead to different decisions. A composite score makes the deployment objective explicit.

Protection profile

Raise γ, δ, λ when trip exposure, recovery, and latency dominate the risk model.

Monitoring profile

Raise α, β, η when reconstruction accuracy and compute budget dominate.

IBR Event Qualification Score

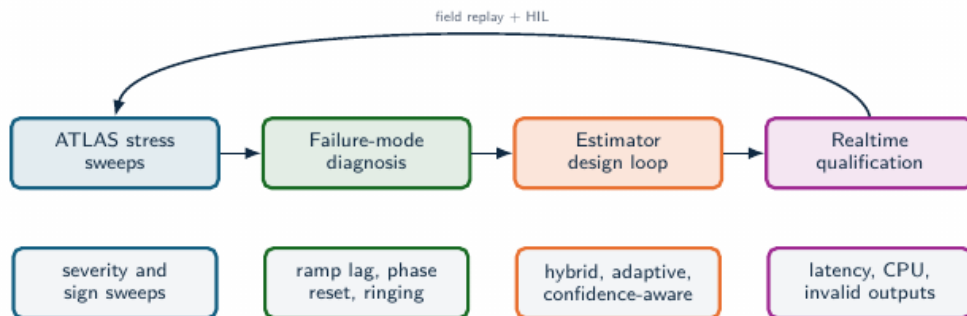
Say:

- When a single number is unavoidable, we define a composite, weighted, event-class score.

→ Finally, let me point to where this work is heading.

Key line: One number when you need it, without hiding the dimensions.

Toward realtime IBR estimators



- The stress analysis identifies which physical stressor breaks each estimator family.
- That diagnosis supports hybrid estimators: event classifiers, harmonic-aware preprocessing, adaptive bandwidth, and self-reported confidence.
- Validation should combine synthetic truth, EMT replay, hardware-in-the-loop, and field oscillography.

Toward realtime IBR estimators

Say:

- Looking ahead: real-time estimators, integrating Simulink microgrid fault records, adding modern and machine-learning methods, C++ acceleration, and co-simulation with ANDES.

→ And with that, let me close.

Key line: This is where the benchmark goes next.

Validation agenda and conclusion

Conclusion

- Field disturbances define the measurement problem; controlled synthetic truth makes estimator behavior explainable.
- The study compares estimators across objectives, regimes, and deployment constraints, including IBR-relevant stress modes.
- Empirical conclusions remain tied to evidence level, stress regime, and replication depth.

Validation agenda

- Full ATLAS sweeps: fixed policy, 100-run Monte Carlo, statistical rank-stability analysis.
- Composite qualification profiles for forensic, PMU-augmentation, and relay-adjacent tasks.
- Three-phase, unbalance, and converter-control interaction coverage.
- Real-event replay from DFR/PMU/oscillography with synchronized ground-truth proxies.
- Estimators that optimize the full objective vector: frequency error, RoCoF, trip exposure, settling, CPU, latency.

Result: estimator choice is tied to event class, metric target, and compute envelope; estimator qualification must declare the stress regime, latency limit, and evidence depth.

Validation agenda and conclusion

Say:

- To conclude: there is no universal frequency estimator.
- The defensible choice depends on the event, the metric, and the compute budget.
- And OpenFreqBench gives the community a reproducible way to make exactly that choice.

Key line: Thank you very much. I would be glad to take your questions.